AHR v2 for 9×9 Go

Teacher learning, bounded tactical reading,
neural search, and empirical results

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Abstract. This note reconstructs the successful AHR v2 Go player from recovered, source-pinned notebooks and experiment records. A shared residual network produces Scout move energies, Worker outcome energies, score estimates, and ownership estimates. Offline teacher learning supplies the network; a fixed Queen scheduler combines batched PUCT search with exact immediate-capture checks and resource-bounded tactical reading. The selected playing model is update 8,016 of a 500,000-position training run. In a 100-game development benchmark it won 31 of 40 games against GNU Go level 10 and none of 60 games against three visit-capped KataGo settings. Mean move time was 2.009 seconds, but the nominal two-second budget was soft. These observations establish a useful functioning Go player under the recorded protocol. They do not establish general AHR superiority, full-game optimality, hard real-time operation, or parity with KataGo.

31 / 40 GNU Go L10	**1,053,799 parameters**	**2.009 s mean move**
77.5% observed wins	500,000 training positions	93.24% exceeded 2.000 s

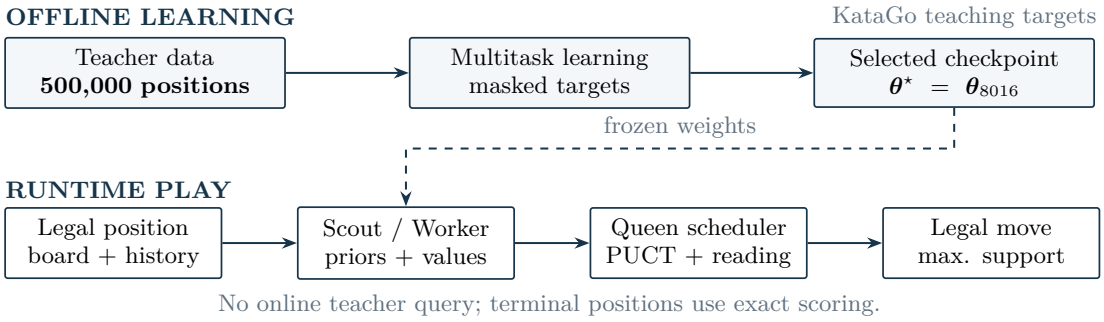


Figure 1: Offline learning and deployed decision making. The external teacher supplies training targets. Runtime moves are chosen by the recovered AHR network, its search, and its rule-based tactical reader; the teacher is not queried to choose the player's moves.

Evidence scope. This is a retrospective research note, not a new training run or a peer-reviewed strength claim. Mathematical details were checked against recovered notebook export hashes. The source bundle contains the evidence used for the plots. The trained checkpoint payload and prepared training arrays were not present in the inspected local recovery copies.

1 State, encoding, and the learned network

Let a position be $s = (B, c, \mathcal{H}, k, r)$, where $B \in \{-1, 0, +1\}^{9\times 9}$ is the board, black is $+1$, c is the current player, $\mathcal{H}$ is the repetition history, $k = 7.5$ is komi, and r counts consecutive passes. Legal actions $\mathcal{A}(s) \subseteq \{0, \ldots, 81\}$ include pass. The transition $T(s, a)$ uses area scoring, forbids suicide, and enforces positional superko. Two consecutive passes terminate the game. Terminal values use exact rules:

$$z(s) = c\ \mathrm{sign}(\mathrm{area}_B - \mathrm{area}_W - k)\,. \tag{1}$$

Full history is retained for legality even though the neural encoder contains only four previous boards. Sources: notebooks 01–02, [E1].

Planes	Contents of $X(s) \in \mathbb{R}^{21\times 9\times 9}$
0–9	Own/opponent stones now and on the previous four boards, always relative to the current player; missing history is zero.
10–15	Own/opponent groups with one, two, or at least three liberties.
16	Legal board-point mask.
17	Constant $-ck/(81 + \lvert k\rvert)$.
18–20	Black-to-move indicator; ones; $\min(r, 2)/2$.

Table 1: Exact input-plane groups. Pass legality is handled in the 82-action mask.

With $\rho = \mathrm{SiLU}$, $G = \mathrm{GroupNorm}_8$, and bias-free 3×3 convolutions C, the width-96 trunk is

$$H_0 = \rho(G(C_0 X)), \tag{2}$$

$$H_{\ell+1} = \rho[H_\ell + G(C_{\ell,2}\rho(G(C_{\ell,1} H_\ell)))]\,, \quad \ell = 0, \ldots, 5. \tag{3}$$

Pooling yields $g = [\mathrm{mean}_x\, H_6(x); \max_x H_6(x)] \in \mathbb{R}^{192}$.

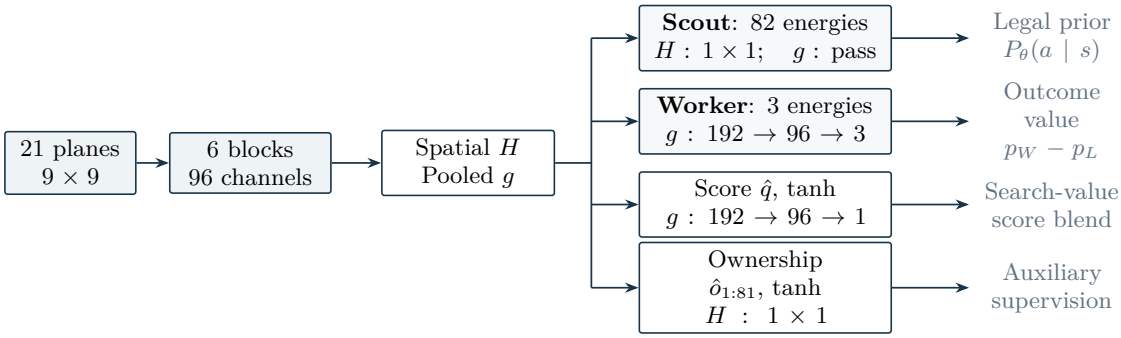


Figure 2: Shared representation and four heads. "Scout" and "Worker" name learned outputs of one network; the bounded tactical reader is a separate rule-based procedure.

The Scout uses a 1×1 board-energy convolution and an affine pass energy. Worker energies use a $192 \to 96 \to 3$ SiLU MLP. Score uses a separate $192 \to 96 \to 1$ MLP with tanh; ownership uses a 1×1 convolution with tanh. Temperatures are both one:

$$P_\theta(a \mid s) = \frac{\mathbf{1}[a \in \mathcal{A}(s)] e^{-E^S_\theta(s,a)}}{\sum_{b\in\mathcal{A}(s)} e^{-E^S_\theta(s,b)}}, \tag{4}$$

$$p^W_\theta(y \mid s) = \mathrm{softmax}_y(-E^W_\theta(s)), \qquad V_\theta(s) = p^W_\theta(\mathrm{win} \mid s) - p^W_\theta(\mathrm{loss} \mid s). \tag{5}$$

2 Teacher data and the successful training objective

The data source is `quintic/go9x9`, pinned to the immutable commit recorded in Appendix A [3]. The converter rejects a whole game unless every move replays legally, every recorded board matches, and the game terminates under AHR's rules. This matters because the publisher permits multi-stone suicide while AHR forbids suicide. At most 32 positions per accepted game are selected by unique integer linspace. The retained bank has 500,000 training positions and 8,192 development positions; both sides' turns are eligible. Sources: notebooks 12–13 and the training records, [E2].

The winning preparation uses *raw* teacher policy and value. Available searched-policy fields are validated but not used as policy targets. For a selected row,

$$\pi_i^T(a) = \frac{\mathbf{1}[a \in \mathcal{A}(s_i)]\pi_i^{\text{raw}}(a)}{\sum_{b\in\mathcal{A}(s_i)} \pi_i^{\text{raw}}(b)}, \tag{6}$$

$$y_i^T = \left((1-\bar{v}_i^T)/2,\ 0,\ (1+\bar{v}_i^T)/2\right), \quad \bar{v}_i^T = \text{clip}(v_i^{\text{raw}}, -1, 1). \tag{7}$$

The raw scalar is already in the player-to-move perspective; it is not multiplied by color again. The soft three-class target does not assert a teacher draw probability.

The normalized score target is $q_i^T = q_i^{\text{raw}}/88.5$, valid only when marked valid, finite, and $|q_i^{\text{raw}}| \le 88.5$. Final ownership is $o_i(x) = c_i O_B(x)$ and is valid only when the publisher's final white score agrees with the independently replayed black score after reversing perspective: $|S_{\text{publisher,white}} + S_{\text{replayed,black}}| \le 10^{-5}$. Define

$$\langle \ell_i \rangle_m = \begin{cases} \sum_i m_i \ell_i / \sum_i m_i, & \sum_i m_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

For $\text{CE}(p,q) = -\sum_j p_j \log q_j$, the actual loss is

$$\boxed{\begin{aligned}\mathcal{L}(\theta) = B^{-1}\sum_i \text{CE}(\pi_i^T, P_\theta(\cdot \mid s_i)) + B^{-1}\sum_i \text{CE}(y_i^T, p_\theta^W(\cdot \mid s_i)) \\ + \tfrac{1}{2}\left\langle (\hat{q}_\theta(s_i) - q_i^T)^2 \right\rangle_{m_q} + \tfrac{1}{4}\left\langle 81^{-1}\|\hat{o}_\theta(s_i) - o_i\|_2^2 \right\rangle_{m_o}.\end{aligned}} \tag{9}$$

Observed-game-outcome cross entropy is computed by the implementation but has coefficient **zero**. There is no learned group-capture target in this pretraining loss.

Setting	Successful run
Optimizer / batch	AdamW; batch 512; global gradient norm clipped at 2.
Learning rate	Peak 5×10^{-4}; 200-step warmup; cosine decay to 10% of peak.
Weight decay / precision	Decoupled 10^{-4}; CUDA BF16 autocast, scaler disabled.
Sampling	Shuffled 16,384-row blocks, shuffled rows inside blocks; independent D_4 transform per sampled row.
Training endpoint	12 epochs; 11,724 updates; 6,000,000 row exposures.
Selected player	**Update 8,016**, selected by GNU Go development comparisons.

Table 2: The eight-hour setting was a cap. Recorded active training-loop time was 507.50 seconds (8.46 minutes), excluding data preparation and the complete evaluation pipeline.

3 Immediate tactics and bounded capture reading

At a nonterminal search leaf, outcome and score estimates are combined as

$$\widetilde{V}(s) = 0.85V_{\theta^\star}(s) + 0.15\tanh(3\hat{q}_{\theta^\star}(s)), \qquad \theta^\star = \theta_{8016}. \tag{10}$$

For each root move a, let $C(s,a)$ be enemy stones captured immediately, and $D(s,a)$ the maximum own-stone capture by one legal next reply. Then

$$u(s,a) = \tanh((C(s,a) - D(s,a))/3), \tag{11}$$

$$P(s,a) = 0.85P^0(s,a) + 0.15\frac{e^{2u(s,a)}}{\sum_{b\in\mathcal{A}(s)} e^{2u(s,b)}}, \quad b_{\text{imm}}(s,a) = 0.35u(s,a). \tag{12}$$

Here P^0 is the legal Scout prior with a 10^{-6} floor and renormalization. At nonroot nodes $P = P^0$ and $b_{\text{imm}} = 0$. All legal actions remain available. Sources: notebooks 08 and 14, [E3].

A capture task (s, G, A, d) asks whether attacker A can force capture of the surviving original defender stones G in at most d plies. If G_0 is the original target, update G monotonically after each transition:

$$G_a = \{x \in G : B_{T(s,a)}(x) = -A\}. \tag{13}$$

A later replacement stone on a captured point does not resurrect the original target. Without resource interruption,

$$R(s,G,A,d) = \begin{cases} 1, & G = \varnothing, \\ 0, & d = 0 \text{ or } s \text{ terminal}, \\ \bigvee_{a\in\mathcal{A}(s)} R(T(s,a), G_a, A, d-1), & c(s) = A, \\ \bigwedge_{a\in\mathcal{A}(s)} R(T(s,a), G_a, A, d-1), & c(s) \neq A. \end{cases} \tag{14}$$

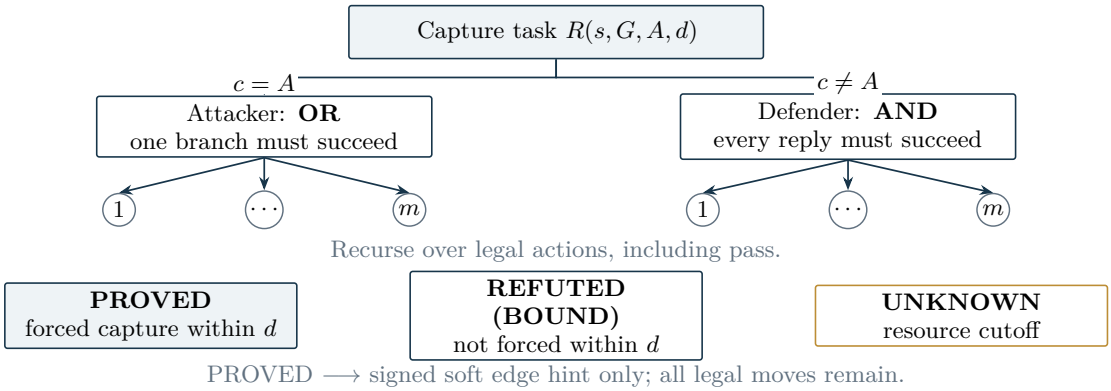


Figure 3: Logical structure of bounded reading. All legal defender replies, including pass, are covered by a proved forced-capture claim. The diagram is schematic, not a recorded Go position.

The reader returns `PROVED_CAPTURE_WITHIN_BOUND`, `REFUTED_WITHIN_BOUND`, `UNKNOWN` on node/deadline exhaustion, or `PENDING` on scheduling-quantum exhaustion. A resolved depth limit is a bounded refutation; it does not establish unrestricted survival.

4 Queen scheduling, batched PUCT, and the played move

The Queen is a fixed scheduler, not a learned controller. Reading targets are groups of at least two stones with at most two liberties, considered at the root and encountered search depths up to two. At the root, task slots are limited to two, leaving room for descendant tasks. Attack tasks target the largest eligible opposing group; candidate-risk tasks inspect own groups after one of the top three prior moves, excluding pass.

Reader limits are six plies, four tasks, 64 entered nonterminal nodes per task, 256 shared nodes, and 16-node scheduling quanta. The 50 ms reader deadline resets per scheduling call. These limits count internal reader nodes, not all transitions or leaves.

Only proved tasks add soft edge-bias increments:

$$b(s,a) = b_{\text{imm}}(s,a) + \sum_{j \rightsquigarrow (s,a)} \sigma_j \, 0.45 \tanh(|G_{0,j}|/4). \tag{15}$$

Here $\sigma_j = +1$ for a proved attacking line and -1 for a candidate exposing its own original group to forced capture. Refuted or UNKNOWN tasks add no hint. Several tasks may contribute to one edge. Immediate-capture witnesses and proved principal lines enter a forced-verification queue.

Each edge stores real visits N, accumulated value W, and temporary reservations ν. With virtual-loss coefficient one,

$$Q_{\text{res}}(s,a) = \begin{cases} (W-\nu)/(N+\nu), & N+\nu > 0, \\ \widetilde{V}(s) - 0.25, & N+\nu = 0, \end{cases} \tag{16}$$

$$U(s,a) = Q_{\text{res}}(s,a) + 1.5 P(s,a) \frac{\sqrt{1+N(s)+\nu(s)}}{1+N(s,a)+\nu(s,a)} + b(s,a). \tag{17}$$

Ordinary traversal maximizes $(U, P, -a)$ lexicographically. The scheduler prioritizes queued verification lines, gathers up to eight distinct nonterminal leaves, evaluates them together, releases reservations, and commits real visits atomically. Backup alternates perspective: along a root-to-leaf path of length ℓ,

$$N(s_j,a_j) \leftarrow N(s_j,a_j) + 1, \qquad W(s_j,a_j) \leftarrow W(s_j,a_j) + (-1)^{\ell-j} v_\ell. \tag{18}$$

Terminal leaves supply exact z; nonterminal leaves supply $\widetilde{V}$. Standard policy/value-guided tree search is an established method [1]; this note describes the recovered implementation and its domain-specific additions.

Verification visits are not policy votes. They still affect ordinary N, W, Q and exploration. Only final policy-support counts subtract them:

$$S(a) = N(s_0,a) - N_{\text{verification}}(s_0,a), \qquad \pi_{\text{search}}(a) = S(a) / \sum_b S(b). \tag{19}$$

When total support is positive, the played move is

$$a^\star = \operatorname*{arg\,lex\,max}_{a \in \mathcal{A}(s_0)} \left(S(a),\ W(s_0,a)/N(s_0,a) + b(s_0,a),\ P(s_0,a),\ -a \right). \tag{20}$$

The second component is $-\infty$ for unvisited edges. With zero total support, the inherited real-visit/prior fallback applies and no valid policy target is emitted. Source: notebook 14, [E3].

5 Evaluation protocol and observed playing results

The selected snapshot is `step-000008016-c19b207e.pt`, from run `ahr-v2-teacher-20261008`. The larger benchmark, `ahr-v2-benchmark-20261009-v1`, used a nominal two-second move budget, a cap of 512 simulations, batch size eight, and root exploration noise disabled. GNU Go 3.8 ran at level 10; KataGo v1.18.1 used a pinned 9×9 teacher network and fresh search caps of 1, 4, or 16 visits. CPU opponents and GPU AHR did *not* receive equal compute or equal wall time.

The 100 games comprise 50 color-swapped pairs: 46 fresh opening families and four empty-board pairs. The 46 nonempty opening families used eight sampled plies; four pairs started from the empty board. Play then followed the pinned rules. The recorded plan excludes training, development, and previously reviewed opening families. Pairing still induces correlation, and later transpositions can occur.

Opponent	Games	Wins	Losses	UNKNOWN	AHR win rate
GNU Go level 10	40	31	9	0	77.5%
KataGo, 1 visit	20	0	20	0	0.0%
KataGo, 4 visits	20	0	20	0	0.0%
KataGo, 16 visits	20	0	20	0	0.0%
All profiles	100	31	69	0	31.0%

Table 3: Completed benchmark 17. Draw count is zero. The aggregate across different opponents is a description of this mixture, not a general playing-strength estimate.

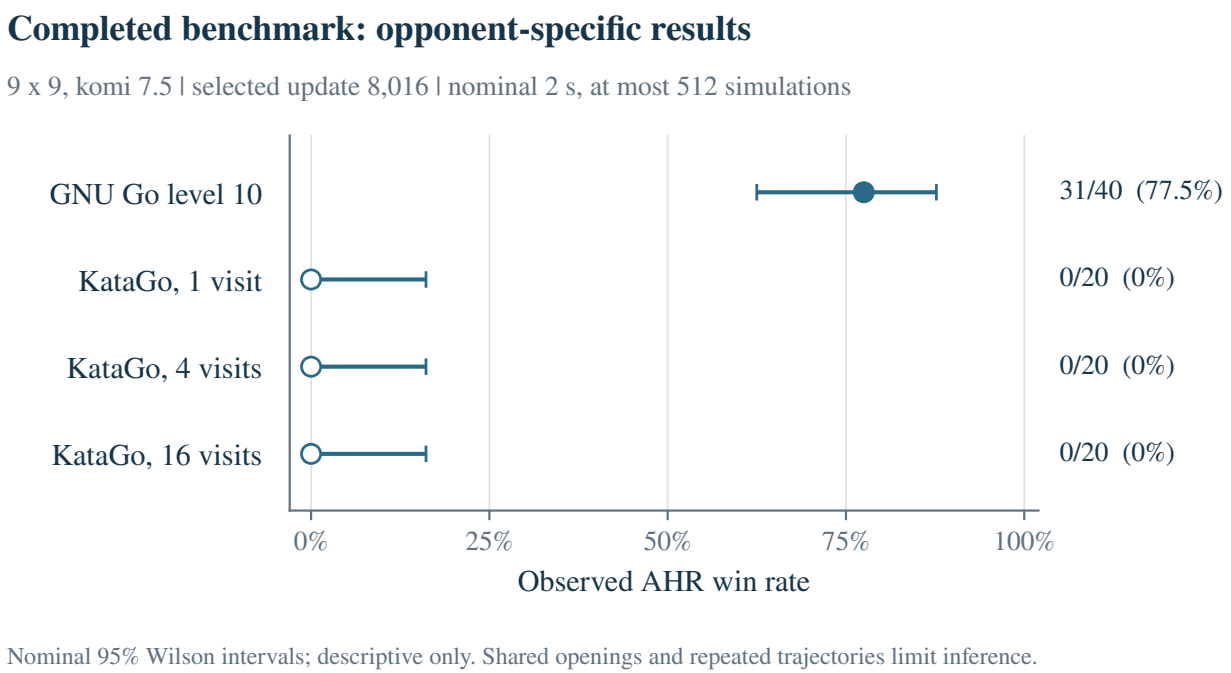


Figure 4: AHR wins by benchmark opponent. Error bars are nominal 95% Wilson binomial intervals, shown descriptively: color-paired openings violate the independent-game assumption. A zero observed win count does not establish zero true win probability.

Provenance limit. The recovered completed worker record directly reports 100 games, 31 wins, 69 losses, no draws, and no UNKNOWN outcomes. Per-opponent counts were recovered from archived diagnostic output that identifies the original game report and its SHA-256. That original benchmark game-trace payload is absent from the inspected copies, so it could not be rehashed or replayed for this note. Sources: [E4]–[E5].

6 Earlier development reviews and checkpoint selection

Before the larger benchmark, the selected player was evaluated with **128 fixed simulations per move and no wall-time limit**. These smaller reviews reported 15/16 wins against GNU Go level 1, 8/8 against level 5, and 8/8 against level 10. They establish the earlier milestone but must not be pooled with the two-second benchmark or interpreted as a controlled improvement between protocols. Sources: original per-game review reports, [E6].

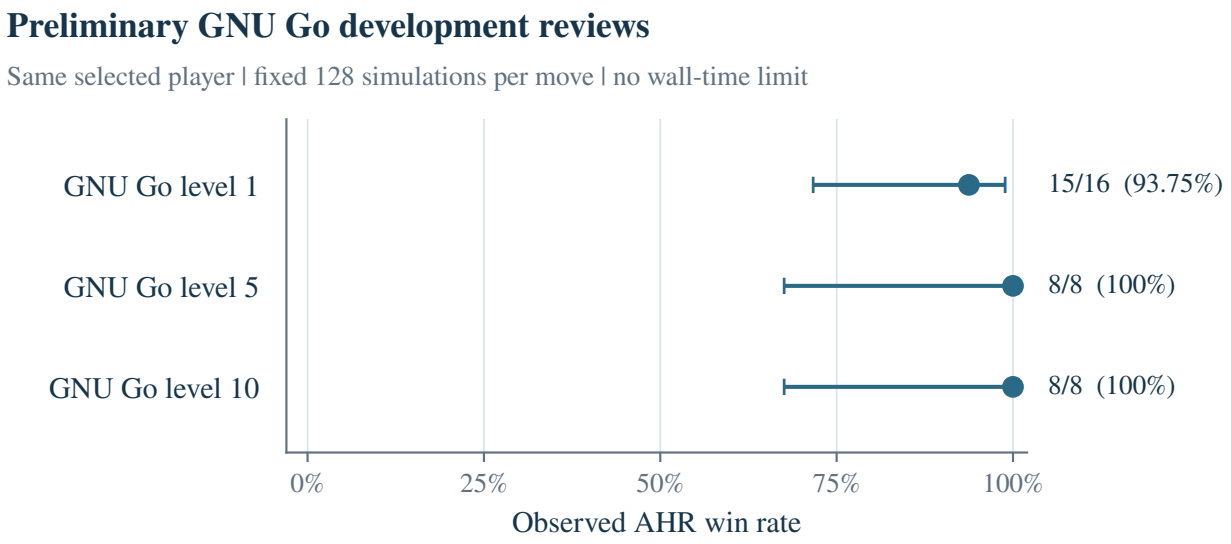


Figure 5: Preliminary development reviews under 128 fixed simulations. Counts are AHR wins. Nominal 95% Wilson intervals are descriptive; the respective records contain only 11, 7, and 8 unique trajectories among 16, 8, and 8 games.

Training loss and playing selection are different decisions. The full run completed 12 epochs and 11,724 updates. Its final 2,048-row development slice recorded policy cross entropy 2.0751, teacher-value MSE 0.1644, score MAE 4.406 points, and joint loss 2.40794. These are final-checkpoint diagnostics, *not* measurements of selected checkpoint 8,016. The observed-outcome metric used 894 valid rows; it does not imply an active game-outcome training loss.

The control record selected checkpoint 8,016 through GNU Go development comparisons. The final update-11,724 snapshot achieved the best recorded development loss but was not the selected player. Small development-match selection can itself overfit; an independent, prespecified confirmation remains necessary for stronger conclusions.

What was demonstrated	What remains unestablished
Winning complete games against GNU Go	Broad human rank, equal-compute superiority, or strong KataGo performance.
Finite-horizon capture reasoning	Unrestricted group survival or full-game optimality.
Functioning teacher learning and search	The causal contribution of each AHR mechanism.
Saved source and checkpoint identities	Availability of the trained weight payload in this local recovery.

Table 4: Scope of the empirical and logical claims.

7 Move latency and the meaning of "real time"

Benchmark 17 recorded 3,564 AHR moves. The mean time was 2.00896 s, the median 2.04026 s, the 95th percentile 2.07431 s, the 99th percentile 2.09594 s, and the maximum 2.22593 s. **3,323 moves (93.24%) exceeded 2.000 s.** These results support interactive play near two seconds per move, not a hard two-second guarantee.

The search checks the time budget between atomic batches; feature extraction, replay checks, reading, and an in-progress batch can overrun the target. The reader's own 50 ms deadline also resets on each scheduling call. The original speed report explicitly describes these as soft targets.

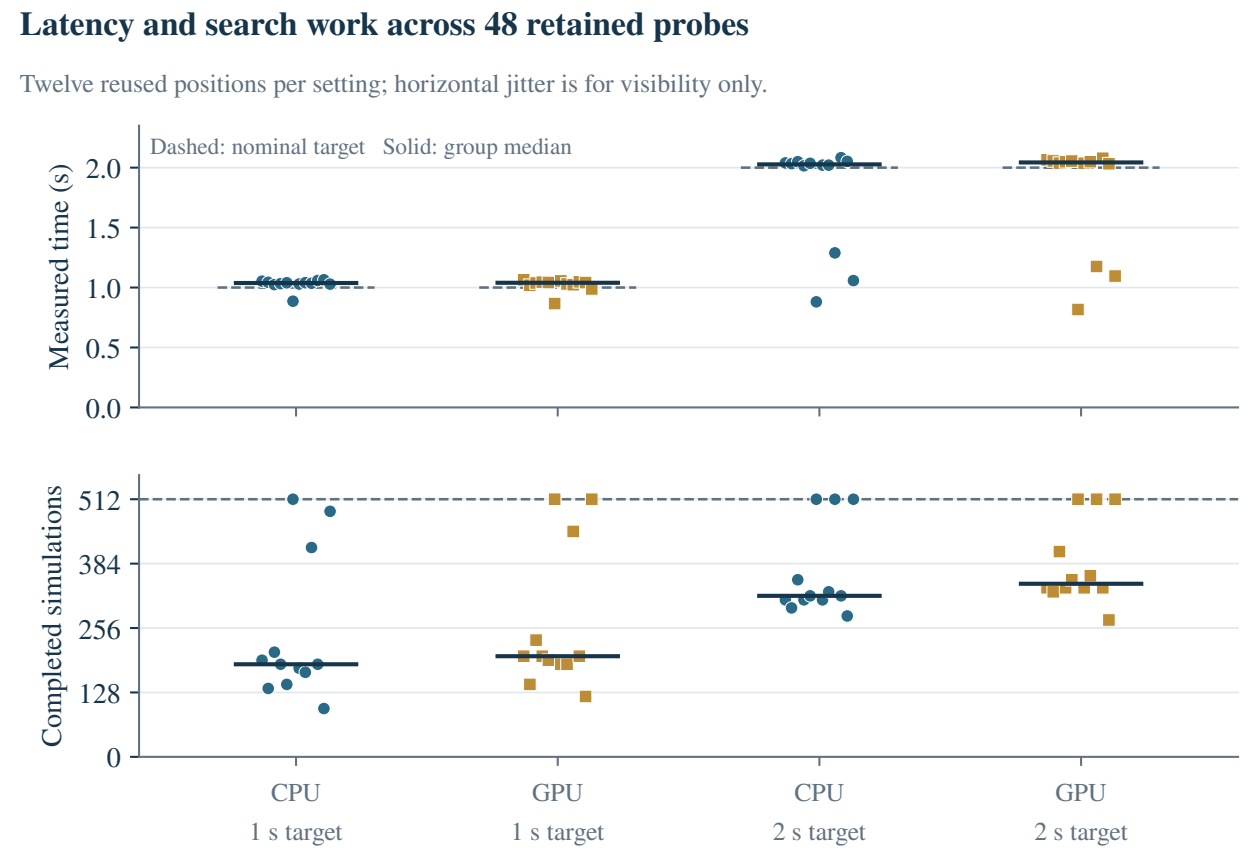


Figure 6: All 48 retained speed probes: 12 positions per CPU/GPU and nominal 1 s/2 s setting. Top: measured latency, with target markers and group medians. Bottom: completed simulations, with the 512-simulation cap. Positions are reused across settings, so these are paired probes, not 48 independent positions. Measurements include replay, features, legality, reading, search, and CUDA synchronization; cold-start transfer/first-forward measurements are separate.

Mean completed simulations in the game benchmark were 347.38 per move; mean policy-support simulations were 339.73, and mean entered reader nodes were 109.10. No move had zero completed simulations. These aggregates describe the recorded workload and do not imply that every move received the same computation.

8 Interpretation, limitations, and next experiments

Practical finding. A compact, teacher-trained neural player with AHR's implemented search machinery exceeded the original observed half-games goal against GNU Go level 1 and achieved 31/40 against level 10 in the later development benchmark. This is the strongest supported practical achievement in the recovered record.

Architectural attribution. The system combines established neural tree-search ideas [1] with domain-specific rule checks, bounded reading, and scheduling. Auxiliary score/ownership supervision also has precedent in Go learning research [2]. Naming network heads Scout and Worker does not itself establish a new learning principle. The existing results do not isolate the benefit of energy parameterization, tactical priors, reading hints, or verification scheduling.

Teacher dependence. Runtime AHR chooses its own moves, but its successful initialization learns from external KataGo-generated policy and value estimates. This result is not evidence for learning the same strength from AHR-only self-play. The teacher's raw estimates, observed outcomes, bounded proofs, and full-game results must remain distinct kinds of evidence.

KataGo gap. There were no wins in 20 games at each of 1-, 4-, and 16-visit KataGo settings. The teacher network used by these opponents is much larger than AHR's compact network, and the budgets differ. The data show a large remaining gap under this protocol; they do not estimate the compute needed to close it.

Statistical scope. Development checkpoint selection, small preliminary samples, repeated trajectories, and color-paired openings limit inference. The plotted Wilson intervals assume independent Bernoulli games and are explicitly descriptive here. The aggregate 31/100 rate is tied to the chosen opponent mixture. No sealed external test, general Elo rating, or human-rank certification is claimed.

Recovery scope. Code and full algorithm state are checkpointed. Exact CPU/CUDA search-continuation fixtures disable reader clocks, establishing equality for node-bounded continuation. Production timed decisions can change with machine load and interruption timing; preservation of state does not make wall-clock-limited search bitwise invariant across machines.

Recovery artifact gap. The inspected local mirrors include code, configurations, summaries, and selected game reports, but no selected `.pt` payload or prepared training arrays. A SHA-256 identifies a payload; it cannot reconstruct its contents. This note reconstructs the mathematical and procedural algorithm, not the missing trained tensor values.

A prespecified next study

Hold the initial weights, data, openings, colors, rules, device, and compute accounting fixed. Compare the complete player against: (i) neural PUCT alone; (ii) PUCT plus immediate tactics; (iii) PUCT plus bounded reading; and (iv) the full verification scheduler. Report wins, UNKNOWN outcomes, capture failures, total node work, and latency percentiles. Use fresh opening families and reserve a confirmation set before selection. Analyze color-paired outcomes at the pair level, rather than treating games as independent. For deployment claims, add cancellation or budget reservation before expensive work and measure actual deadline compliance.

The later notebook-21 attempt produced 1,024 labels across 32 games, but its intended full training did not begin before resource waits and a reboot. It supplies no new playing-strength result and is not included in the plots above.

A Exact run identity and reproducibility details

Field	Recorded identity
Training run	`ahr-v2-teacher-20261008`
Selected snapshot	`playing_candidates/step-000008016-c19b207e.pt`
Model parameters	1,053,799; width 96; six residual blocks.
Training seed	260810; sampler generator seed $260810 + 101$.
Development sampler	Separate generator seed $260810 + 303$; fixed 2,048-row metric slice from the 8,192-row development bank.
Dataset bank	500,000 train positions (15,632 games; 7,562 families); 8,192 development positions (256 games; 119 families).
Category quotas	62,500 training positions and 1,024 development positions per each of eight opponent strata.
Benchmark run	`ahr-v2-benchmark-20261009-v1`; seed 2026100917.
Benchmark search	Soft 2 s; at most 512 simulations; batch 8; PUCT 1.5; virtual loss 1; first-play reduction 0.25; maximum tree depth 256.
Reader	Six plies; four jobs; 64 internal nodes/job; 256 shared; quantum 16; depth scope 2; 50 ms per scheduling call.

Table 5: Values from the pinned source and saved successful run configuration.

For zero-based scheduler step t, the exact learning rate is $5 \times 10^{-4} f(t)$, where

$$f(t) = \begin{cases} (t+1)/200, & t < 200, \\ 0.1 + 0.45[1 + \cos(\pi z_t)], & t \geq 200, \end{cases} \qquad z_t = \operatorname{clip}\left(\frac{t - 200}{11724 - 200}, 0, 1\right). \tag{21}$$

Epoch sampling first permutes contiguous 16,384-row blocks and then independently permutes rows inside each block. The private CPU generator also supplies per-row D_4 transformations. This differs from a uniform permutation of all 500,000 rows.

Opening families are hashes of the lexicographically minimum D_4 transform of the first eight actions. The first eight hexadecimal hash digits modulo 100 assign buckets 0–4 to test, 5–9 to validation, and 10–99 to training. Test payloads are not opened by the training preparation.

Selected checkpoint SHA-256

`e7d1126c4ca962adb6eebcf5754341c4e6f1`
`f3eb8396549e0fb8ee11e47fc1aa`

Selected model identity

`0e8ba8c5405815f7499945d4b83f3020a2`
`ecbedf403227e8d532260da5b1df71`

Recorded original benchmark-report SHA-256

`ccfb38e80b83925256f0d451e8a6dfe81`
`ef3394c6a816a75e446230b1c7f0c5b`

Dataset commit

`4f579b0a21c456f3bb568174d384056351124cd5`

B Evidence inventory and references

All artifact paths below are relative to the accompanying LaTeX project. The file `evidence/source_manifest.json` records the original local recovery paths, copied-file hashes, and notebook export hashes. The report itself does not assert access to the stopped VM or to missing checkpoint tensors.

E1 `evidence/notebooks/01_rules_and_replay.ipynb` and `02_scout_worker_network.ipynb`: state, rules, encoder, architecture, heads.

E2 Notebooks 12–13, copied training/pipeline configurations, and `evidence/training_status.json`, `training_control.json`, `production_recovery.json`: targets, masks, loss, optimizer, selection, inventory, and parameter count.

E3 Notebooks 03, 08, and 14: tree search, immediate tactics, bounded reading, scheduling, backups, and policy-support accounting.

E4 Notebook 17, `benchmark_config.json`, `benchmark_plan.json`, `benchmark_worker.json`, and `speed.json`: protocol, model/opponent pins, completed aggregate, and 48 raw timing probes.

E5 `benchmark_breakdown_recovered.json`: exact retained diagnostic fragment, parsed profile counts, original report hash, and transcript timestamp/ordinal. Recovered from an 11 October 2026 command output; it is not a substitute for the missing full game trace.

E6 `preliminary_gnu1.json`, `preliminary_gnu5.json`, and `preliminary_gnu10.json`: original smaller development reports with per-game records.

Notebook export	SHA-256 prefix (full value in manifest)
02: network	`4dc091ae17169ec3`
03: batched search	`819dd5cdc7c47ee3`
08: tactical search	`8a2f8ad6a6afc77b`
12: teacher preparation	`aa10282ac29c59e8`
13: teacher training	`4c145a1e5312973e`
14: bounded reading	`3681d59ac82497fe`

Table 6: Recovered executable definition hashes matched the recorded run pins.

Plot reproduction. Run `analysis.ipynb` top-to-bottom from this project directory. It reads only the copied evidence, recomputes counts and nominal Wilson intervals, checks aggregate reconciliation, and regenerates all statistical plots. It performs no model inference or training. Diagram sources are editable TikZ files under `figures/`. Compile `main.tex` to regenerate this note. No empirical learning curve is shown because a complete training-metric series was not recovered for this document.

References

[1] D. Silver et al. *Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm.* 2017. arXiv:1712.01815.

[2] D. J. Wu. *Accelerating Self-Play Learning in Go.* 2019. arXiv:1902.10565.

[3] quintic. *Go 9×9: fixed-teacher, mixed-strength games.* Hugging Face dataset, pinned commit recorded in Section 2. https://huggingface.co/datasets/quintic/go9x9.